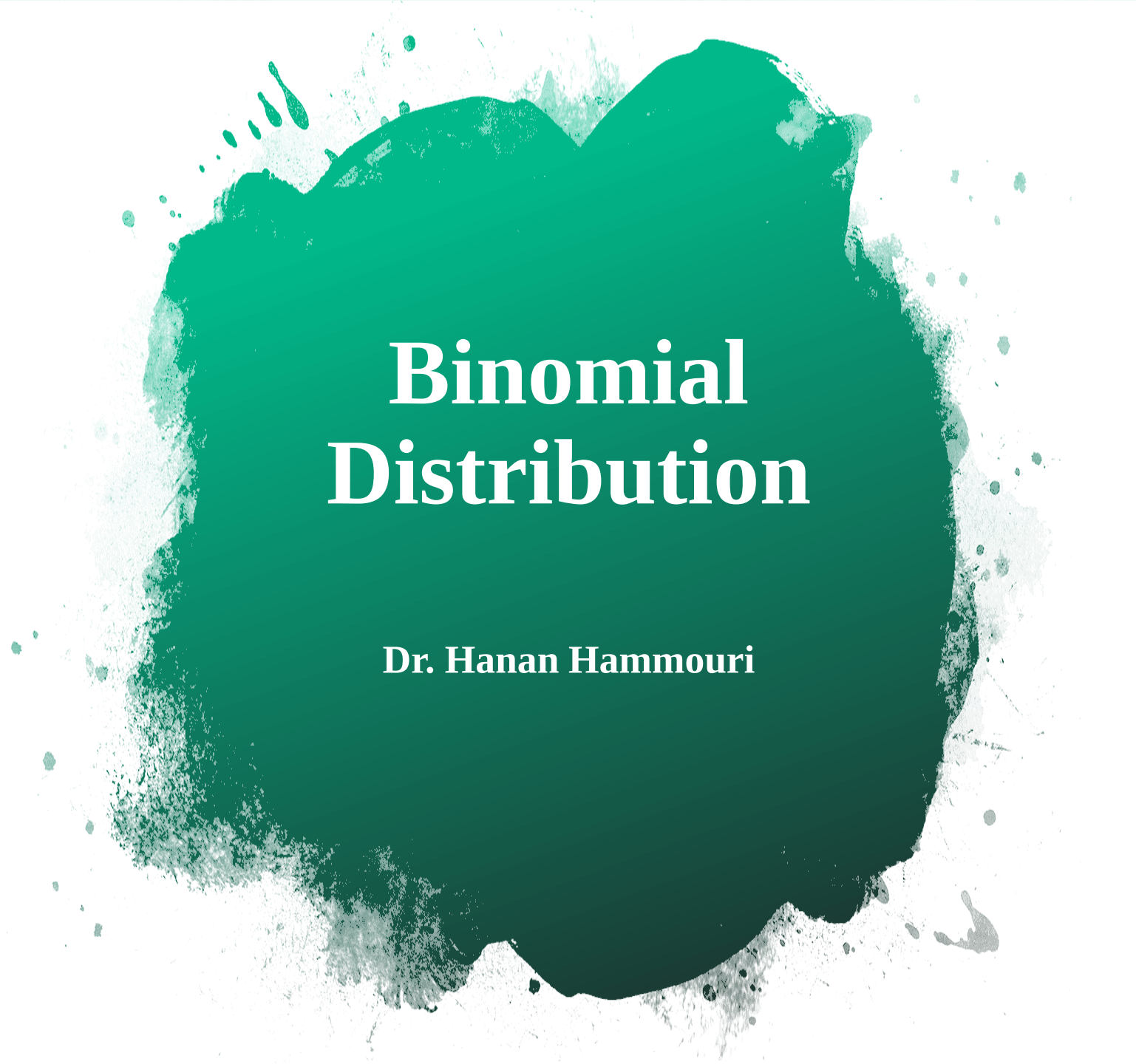




Binomial Distribution

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Binomial Distribution

Definition

A random experiment consists of n Bernoulli trials such that

- (1) The trials are independent
- (2) Each trial results in only two possible outcomes, labeled as “success” and “failure”
- (3) The probability of a success in each trial, denoted as p , remains constant

The random variable X that equals the number of trials that result in a success has a **binomial random variable** with parameters $0 < p < 1$ and $n = 1, 2, \dots$. The probability mass function of X is

$$f(x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad x = 0, 1, \dots, n$$

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Mean and Variance

If X is a binomial random variable with parameters p and n ,

$$\mu = E(X) = np \quad \text{and} \quad \sigma^2 = V(X) = np(1 - p)$$

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Random experiments and random variables

1. Flip a coin 10 times. Let X = number of heads obtained.
2. A worn machine tool produces 1% defective parts. Let X = number of defective parts in the next 25 parts produced.
3. Each sample of air has a 10% chance of containing a particular rare molecule. Let X = the number of air samples that contain the rare molecule in the next 18 samples analyzed.
4. Of all bits transmitted through a digital transmission channel, 10% are received in error. Let X = the number of bits in error in the next five bits transmitted.

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Random experiments and random variables

5. A multiple choice test contains 10 questions, each with four choices, and you guess at each question. Let X = the number of questions answered correctly.
6. In the next 20 births at a hospital, let X = the number of female births.
7. Of all patients suffering a particular illness, 35% experience improvement from a particular medication. In the next 100 patients administered the medication, let X = the number of patients who experience improvement.

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Example

Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant. Find the probability that in the next 18 samples, exactly 2 contain the pollutant.

Let X = the number of samples that contain the pollutant in the next 18 samples analyzed. Then X is a binomial random variable with $p = 0.1$ and $n = 18$. Therefore,

$$P(X = 2) = \binom{18}{2} (0.1)^2 (0.9)^{16}$$

Now $\binom{18}{2} = 18!/[2! 16!] = 18(17)/2 = 153$. Therefore,

$$P(X = 2) = 153(0.1)^2(0.9)^{16} = 0.284$$

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Example

Determine the probability that at least four samples contain the pollutant. The requested probability is

$$P(X \geq 4) = \sum_{x=4}^{18} \binom{18}{x} (0.1)^x (0.9)^{18-x}$$

However, it is easier to use the complementary event,

$$\begin{aligned} P(X \geq 4) &= 1 - P(X < 4) = 1 - \sum_{x=0}^3 \binom{18}{x} (0.1)^x (0.9)^{18-x} \\ &= 1 - [0.150 + 0.300 + 0.284 + 0.168] = 0.098 \end{aligned}$$

Determine the probability that $3 \leq X < 7$. Now

$$\begin{aligned} P(3 \leq X < 7) &= \sum_{x=3}^6 \binom{18}{x} (0.1)^x (0.9)^{18-x} \\ &= 0.168 + 0.070 + 0.022 + 0.005 \\ &= 0.265 \end{aligned}$$

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Example

For the number of transmitted bits received in error , $n = 4$ and $p = 0.1$, so

$$E(X) = 4(0.1) = 0.4 \quad \text{and} \quad V(X) = 4(0.1)(0.9) = 0.36$$

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You're a telemarketer selling service contracts for a certain store. You've sold 20 in your last 100 calls ($p = .20$). If you call **12** people tonight, what's the probability of

- A. No sales?
- B. Exactly 2 sales?
- C. At most 2 sales?
- D. At least 2 sales?

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$$n = 12, p = .20$$

$$\text{A. } p(0) = .0687$$

$$\text{B. } p(2) = .2835$$

$$\begin{aligned}\text{C. } p(\text{at most } 2) &= p(0) + p(1) + p(2) \\ &= .0687 + .2062 + .2835 \\ &= .5584\end{aligned}$$

$$\begin{aligned}\text{D. } p(\text{at least } 2) &= p(2) + p(3) + \dots + p(12) \\ &= 1 - [p(0) + p(1)] \\ &= 1 - .0687 - .2062 \\ &= .7251\end{aligned}$$